

BenQ monitors in India

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Flexible displays

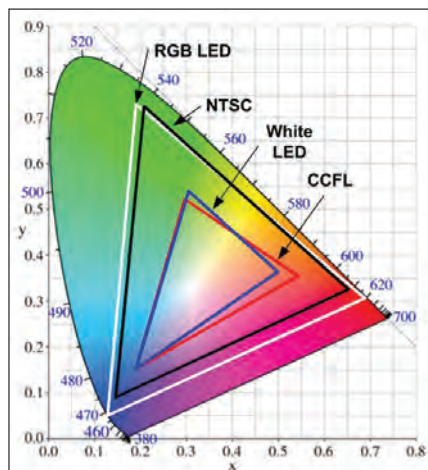
LG showcased a 11.5-inch sheet of flexible e-paper

phosphor formulations used on the tubes used for general lighting. Recently CCFLs have appeared with modified phosphors to give a greater display gamut, for example Sony's Wide Color Gamut - CCFL (WGC-CCFL) with a claimed gamut 30 per cent larger than a conventional CCFL, but it looks like these are going to be eclipsed by the new RGB LED back-lights, at least in part because using LED back-lights reduces the amount of mercury released into the environment.

By comparing the spectral output graphs of the two types of back-light it is easy to see that RGB LED produces higher peaks for red, green and blue and covers more of the visible spectrum. This results in a considerable improvement in gamut can be seen from looking at the gamut area plots.

RGB back-lights save the planet

RGB LED back-lights offer great advantages in terms of power saving and a longer service life than CCFL. According to Sharp, for example, their RGB displays consume 40 per cent less power and offer a back-light service life of 100,000 hours, about two thirds longer than regular LCD



This diagram shows the various gamuts of different display types plotted as 2D areas using the XYZ colour space

TVs. To put this in perspective - if the display is turned on for 5 hours a day, this is 20,000 days or over 57 years.

Back-light layout

The older panel designs with fluorescent back-lights normally use edge lighting where a CCFL is mounted near one edge of the display. Light from the lamp is guided and spread across the back of the display using a narrow wedge shaped light-guide. The light is further disbursed using a translucent diffusion panel and a prismatic panel that helps reflected light to bounce around. Some displays may use two CCFLs top and bottom, or even on large displays a number

of tubes may be mounted in parallel behind the panel.

There are several ways to arrange and drive the LEDs in a LED back-light and these are known as 0D, 1D and 2D.

0D simply uses a number of LEDs and a diffuser system to evenly back-light the entire LCD panel, a direct replacement of the CCFL without any added functionality.

In a 1D back-light the LEDs are arranged in horizontal rows behind the panel and can be brightness modulated to achieve a crude degree of contrast enhancement.

Lastly, in a 2D back-light the LEDs can be arranged in groups (for an RGB back-light also in sub-groups of red, green and blue) that can be individually brightness modulated, an improvement in contrast over different areas of the display (as shown in the two photographs of an RGB back lit television display).

Colour temperature

Previously the only way to adjust display colour temperature was to adjust the individual gain of the red, green and blue colour channels driving the display. RGB back-lights provide the opportunity to change the colour temperature of the back-light itself. In other words, the shade of the



Two photographs of identical LCD back lit televisions driven with the same signal. On the left the LCD as well as the modulated 2D back-light is being driven with the image signal while on the right only the LED back-light is brightness modulated

'white' light provided by the back-light can easily be varied, for example from a 'warmer' reddish white - 5,000K, to a 'cooler' blue-white - 9,300K.

Contrast improvement using LED back-lights

LCD panel contrast can be hugely improved by turning the back-light off. In this case the idea of representing contrast as a ratio gets a bit silly because the ratio of - some light - to - no light - is infinite. However this really only applies in a completely dark room. In practice there is often a high level of ambient light, which is reflected by the internal structure of the display, degrading contrast.

LED back-lights can be turned on and off very rapidly and they can also be brightness modulated over a wide range, neither of which can easily be done with CCFL

back-lights. LED brightness modulation is usually performed by driving them with a square wave and modulating the duty ratio of the wave (pulse width modulation). This technique does introduce its own problems of RF radiation and the difficulty of actually transitioning to a true off state.

Of course, if a small high resolution panel could be made with a set of RGB LEDs at every pixel, the LCD panel would not be needed since the image could be reproduced entirely by brightness modulated LEDs. Large LED-only displays, where each group of three LEDs is small compared to the overall display size, do exist and are used for applications such as advertising hoardings. However, at the moment it isn't practical to produce smaller high resolution displays in this way, so the LCD panel is retained to provide the required resolution and LED back-lights are used to enhance the contrast (and improve the colour gamut). This is achieved by dividing the display up into quite low resolution segments, typically around 128 segments may be used. Within each of these segments, contrast may be enhanced by modulating the brightness of all the LEDs in that segment. This does mean there is some light spill or haloing in the areas where the image on the LCD

changes abruptly from light to dark. Using more segments will result in smoother contrast changes across an image. At the moment, the marketing departments of most display manufacturers are keeping quite about the exact details of their RGB backlit displays, Sony for example will not say how many LEDs they use or in how many segments. Sony do say they use 2 green with 1 red and 1 blue LED in their LED clusters. This compensates for the generally lower light output of green LEDs compared to red and blue.

With computer monitors that are used primarily to display business graphics its debatable that there is any advantage in using the area based back-light dimming that is being employed in the current crop of LED back-light TV's for contrast enhancement. This really only comes in to its own for photo